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Sentiment Analysis and Machine Learning-Based Stock Prediction for 10 KLSE Index Stocks

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Abstract: Research Question: How can the integration of localized investor sentiment from social media with historical stock data enhance the predictive accuracy of machine learning models for Kuala Lumpur Stock Exchange (KLSE) index stocks? **Motivation:** This study introduces novelty by utilizing a localized sentiment dataset extracted from the KLSE Screener, a platform specific to Malaysian investors. This approach better reflects regional investor behavior, language diversity, and local market dynamics compared to global datasets, offering a more contextualized analysis of the Malaysian financial ecosystem. **Idea:** The core idea is to develop a hybrid prediction model that integrates historical stock data with quantified investor sentiment scores. The central hypothesis is that this integration will yield superior forecasting performance compared to models using historical data alone. The dependent variable is the stock's closing price, and key independent variables include the historical price features and the sentiment score. **Data:** Ten years of historical price data were analyzed for the top KLSE stocks. This was combined with one year of investor comments scraped from the KLSE Screener. Sentiment was measured using the Valence Aware Dictionary and Sentiment Reasoner (VADER) lexicon tool. **Method/Tools:** After preprocessing and normalizing the data, six machine learning models implemented and compared. Model performance was evaluated using several performance indicators. **Findings:** XGBoost emerged as the best-performing model, MSE=0.0001 and R^2=0.99998, effectively capturing complex patterns between price and sentiment data. Correlation analysis revealed mixed, generally weak relationships between sentiment and price movements across different stocks, indicating that investor sentiment in this context is often reactive rather than predictive. This finding adds nuance to literature that often assumes a straightforward positive correlation. **Contributions:** This paper's primary contribution is twofold: it demonstrates the superiority of ensemble methods like XGBoost for financial forecasting with sentiment data, and it provides a pioneering analysis using a localized Malaysian sentiment dataset, establishing a foundation for more regionally-aware financial market research.

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1. Introduction

In Malaysia, the stock exchange market is called Bursa Malaysia, also known as the Kuala Lumpur Stock Exchange (KLSE). It is a platform where investors can purchase and sell stocks of listed companies. Stocks can be easily traded in the stock exchange market through brokerages and electronic trading platforms. Most stock traders aim to buy stocks at the lowest price, expecting them to increase in value, and then sell them at a higher price to gain a profit. If traders or investors can predict stock trends and patterns, they can earn a considerable return on their investments. Stock market prediction helps investors make informed decisions and minimize risks to maximize investment returns. However, determining stock market trends is challenging due to the inherent volatility of the market. According to Koukaras *et al.* (2022), stock prediction approaches are generally categorized into four groups: fundamental analysis, technical analysis, machine learning, and sentiment analysis. The emergence of data mining and machine learning techniques is widely used to recognize hidden patterns in the stock market, while sentiment analysis helps understand public sentiment and behaviour toward the stock market.

Dynamic, unpredictable, non-linear, and volatile is the nature of stock markets. Predicting stock price movement was challenging because of the various factors that may affect stock price trends. The comparison of fundamental analysis and technical analysis used to predict the stock price movement does not give a very satisfying result due to the lack of multiple information, while the comments and reviews of the stock investor or trader on the social media platform have been ignored. To predict the stock price, there are many machine learning algorithms have been used, such as decision tree, support vector machine, random forest, neural network, and many other algorithms that have been developed to forecast the stock market. However, the model of algorithms introduced provides a noncognitive result that does not represent a human behaviour or its expression on the stock market. The research, such as Gifty and Li (2024), stock predicting using different algorithms like Autoregressive Integrated Moving Average (ARIMA), Extreme Gradient Boosting (XGBOOST), Long Short-Term Memory (LSTM), and the research from Modi et al (2019) using a support vector machine to do prediction. All the analysts focus on the historical data to simulate the probabilistic prediction without considering the information of sentiment analysis from public opinion. A failure to capture the sentiment data during the stock price prediction has caused the investor to be unaware of rapid shifts in the stock market. With the integration of public opinion and historical data on the stock market, the accuracy of predictions on the stock market will improve significantly. While many prior studies have applied sentiment analysis to the financial market, but rely on a global dataset that uses Twitter or news headlines that do not represent the local market. Compared with other existing research that only focuses on using global datasets, this study localized public investor sentiment, which was extracted from the KLSE screener that referring to Malaysian regional investor behaviour, language diversity, and local market dynamics, which better reflect the sentiment landscape of Malaysian retail investors. By leveraging historical data and market sentiment with localized sentiment data enhances the predictive accuracy of the stock price forecasting model, which uses advanced algorithms to capture complex structure data and the nonlinear relationship between market sentiment and price movement. Additionally, this study also offers insights into economic conditions by reflecting capital flow, business investments, and behavioural patterns within Malaysia's financial ecosystem. Enhancing stock prediction techniques can help mitigate market volatility and uncertainty, boosting investor confidence and contributing to Malaysia's economic and capital market growth.

Thus, this study tends to leverage the valuable sentiment information that can be retrieved from social media, which provides investor behaviour by considering the historical trend of the stock market for prediction, with a focus on Malaysia's top 10 KLCI index stocks. The primary objective of this research is to: (1) develop a machine learning model that integrates with sentiment analysis to predict the stock market trend using historical data and sentiment data from KLSE screener user comments; (2) compare the error of various ML models including Multiple Linear Regression, Random Forest, Support Vector Regression, Extreme Gradient Boost, Multilayer Perceptron Neural Network and Long Short-Term Memory; (3) determine the correlation between the predicted of stock market trend and sentiment analysis is related to each other.

2. Literature Review for Theoretical Background

2.1 Machine Learning Approach for Stock Market Prediction

Machine Learning has been widely used to study and explore the prediction of stock price direction. The research by Al Ridhawi (2021) has explained the machine learning models in three phases: training of the algorithms, the validation of the result, and the testing of the algorithms used to determine the algorithms' performance. Machine learning is usually classified as supervised and unsupervised learning. According to Ashfaq *et al.* (2017), these techniques are used to collect training data in which instances are marked with labels, and each label shows a specific instance class. It includes the process of providing a machine learning model with a training dataset that contains sample input and output during training. The model learns by estimating a function that maps the input to the output. This allows the model to estimate the output for input that has not been seen during training. Some examples of supervised learning techniques are Linear Regression, Support Vector Machine (SVM), Random Forest, K-Nearest Neighbour, and Decision Tree are models used to forecast stock market prices and patterns by providing useful historical price analysis.

A prior study from Ashwini and Sakshi (2020) attempted to predict the National Stock Exchange from the Indian stock market by using machine learning algorithms like Random Forest, Support Vector Machine, KNN, and Logistics Regression. Among the prediction models, Random Forest gives the highest accuracy rate for the prediction, which reaches 80.7% and a recall rate of 78.3% as compared with another model.

In the previous discussion, several machine learning algorithms have been used for stock trend prediction, but not all the models are efficient to do predictive analysis. To enhance the predictive model for stock movement, different techniques have been built that integrate different models together to get better accuracy performance. A comparative study by Nabipour *et al.* (2020) has integrated machine learning and deep learning algorithms to predict the stock market trend using continuous and binary data. The overall result in this study shows that the whole algorithms predict well as they trained with the continuous data with up the 67% accuracy, but the model performance is improved when they are trained with the binary data with up to 83% accuracy, and deep learning algorithms were the best superior models when including the binary data.

The study from Mokhtari *et al.* (2021) has developed a stock market prediction model based on technical analysis and fundamental analysis. The technical analysis approach was based on historical price data, and the fundamental analysis focused on the classification of the sentiment based on public news. In this research, the technical approach was integrated with the machine learning model to predict the stock price with features such as open price, close price, low price, high price, volume, moving average, MACD, and RSI. In contrast, fundamental analysis includes the features in the Twitter text and financial content that provide sentiment to the stock market. The result of this research shows that the linear regression model predicts the closing price and shows a shallow error value in the technical analysis, while the fundamental analysis, which includes the sentiment analysis, the Support Vector Machine (SVM) model can predict the public sentiment with an accuracy of 76%. This research examines the effectiveness of machine learning in both technical and fundamental approaches that are most used to predict the stock market using historical data and public sentiment expression, where positive sentiment refers to price increases and negative sentiment refers to price drops.

Low *et al.* (2017) have trained the Naïve Bayes machine learning model to predict the stock price using historical data and sentiment data. The use of the sentiment score has been examined to check the relationship between the stock prices. The result of the correlation between the stock price and news sentiment regarding Genting Berhad in this research has shown that there is a positive correlation between each other. This means that the news sentiment will affect the market mood, which will also affect the stock price movement.

2.2 Sentiment Analysis in the Financial Market

Sentiment analysis (SA), or opinion mining, extracts and examines public mood using natural language processing (NLP) (Md Nohh *et al.*, 2021). It classifies user reviews, comments, and social media opinions as positive, negative, or neutral. Pang and Lee (2004) divided sentiment analysis into two tasks: detecting sentiment in text and determining its polarity and strength. A lexicon-based approach identifies emotional states (Ali *et al.*, 2022). SA plays a crucial role in marketing, customer service, risk management, and decision-making. It is also widely applied in stock market prediction, helping investors understand market sentiment and trends.

The research by Koukaras *et al.* (2022) examines the multiple integrated sentiment analyses, such as TextBlob and VADER, from Twitter with a machine learning model. The result of this study shows that analyzed using a Valence Aware Dictionary Sentiment Reasoner (VADER) and Support Vector Machine (SVM) is the best result was obtained with an F-score of 76.3% and Area Under Curve (AUC) of 67% suggesting that this model can forecast the stock trend. Consequently, sentiment tools such as TextBlob and VADER will influence the accuracy of the stock predictions, which, in this research, shows that VADER is the best tool to integrate with historical data for predictions.

Albahli *et al.* (2022) have proposed a novel StockSentiWordNet (SSWN) model, which extends the standard opinion lexicon named SentiWordNet (SWN) through the terms specifically related to the stock market to train the extreme learning machine and recurrent neural network for stock price predictions. The finding shows that the model outperforms which achieving an accuracy of 86.06%. From the research, sentiment plays a sensitive form of analysis that relies on a dictionary and library to compute the sentiment compound score. A reliable sentiment score can be used to enhance the accuracy of the machine learning model for stock price predictions.

However, some research has argued that the use of sentiment analysis is dependent on the source of the data used for analysis. De Lange (2023) has tested various machine learning models, such as Neural Networks, Support Vector Machines, Naïve Bayes, Long Short-Term Memory, and Random Forest, to predict the stock market movement. The result of this study shows that the model could predict directional stock movement with an average accuracy of 56.4% while with added sentiment data, an increment of 6.07% was achieved. These results indicate that sentiment can increase the efficiency of the price prediction, but to what extent the increased accuracy can contribute to the stock prediction market.

The existence of numerous studies on stock market prediction using sentiment analysis has demonstrated the influence of the news and social media on the stock market. However, the review of the literature explaining the sentiment analysis relies on the source of data, which means that more interpretable sentiment data will represent the mood and behavior of the stock market environment. So, this study would like to expand the study by integrating the historical data and the sentiment analysis data based in Malaysia, which would represent the different cultures of Malaysia that have diversified languages used in the sentiment model.

3. Methodology

3.1 Data Sources

This study dataset was extracted from two sources. One is from the stock historical dataset extracted from the website investing.com from 28-10-2014 to 28-10-2024, which is for 10-year record for the information of open price for stock, closing price on particular trading day, the highest and lowest price recorded, and volume of the quantity of stock on particular trading days. Overall, the period includes 2450 trading days, excluding the days when the stock market is closed. The selected stock is from the KLCI index, composed of the 30 largest companies in Bursa Malaysia. The companies are CIMB Holding, Public Bank, Maybank, Petronas Chemical, CelcomDigi, Press Aluminium Holding, Tenaga Nasional Berhad, YTL Corporation Berhad, Genting Berhad, and Nestle. The second dataset, the sentiment data, was extracted from the KLSE Screener browser, which is klsescreener.com. The review and comment were extracted using the Instant Data Scrapper. The dataset total comprises 7686 reviews and was extracted from the past 1 year, 28/10/2023 to 28/10/2024, obtained from the KLSE screener user review based on the selected company stock and the quantity of reviews depends on user comments posted on that 1-year record. Table 1 outlines the attributes, description of data, and data types used in this study.

Table 1: Data dictionary and data type

Attribute	Description	Data Type
Date	The date specified for each trading day is in the format of month-day-year.	Date
Open Price	The first traded price of the day on a given market is open for trading.	Float
Closing Price	The last price that the stock trades during the regular market hours is closed for trading.	Float
Highest Price	The highest price of the stock is reached during the market hours.	Float
Lowest Price	The lowest price of the stock is reached during the market hours.	Float
Volume	The number of shares traded at the end of market hours.	Float
Username	The username is recorded when a review or comment is made at the KLSE Screener application.	Text
Liked Comment	The total number of likes on the comment	Integer
Review/Comment	The user's opinion or expression is written on the KLSE Screener application	Text
Date	The date that the user wrote their comment on the KLSE Screener application	Date

3.2 Data Preprocessing for Sentiment Data

Textual data collected from the KLSE screener is more complex and unstructured in the form of text, symbols, emojis, and others. Due to the unstructured form of textual data, text preprocessing is required before conducting the sentiment analysis. To conduct the preprocessing and sentiment analysis, a Python programming language script was used to run the data preprocessing and sentiment analysis. They used Python along with the libraries, including Pandas and Natural Language Toolkit (NLTK), to clean the data. The process of data preprocessing includes noise removal, tokenization, stopword removal, stemming, and lemmatization.

3.3 Sentiment Analysis (VADER)

This study applied the sentiment identification of the review dataset by using the lexicon rule-based algorithms of the Valence Aware Dictionary for Sentiment Reasoning (VADER). A sentiment analysis consists of the sentiment content in user reviews, comments, or opinions in social media, which focuses on classifying the text based on the polarity of the text can be either positive, negative, or neutral. In the research by Maqbool *et al.* (2023), VADER has applied sentiment analysis by VADER on the financial news to calculate the sentiment score to predict the stock prices. In VADER, the analyser generates the score of negativity, neutrality and positivity of the sentences and calculates the compound score based on the earlier score. Compounds correspond to the sum of the valence score of each word in the lexicon and determine the degree of sentiment. As shown in Figure 1, the review was classified compound score of positive (> 0), negative (< 0), and neutral (=0).

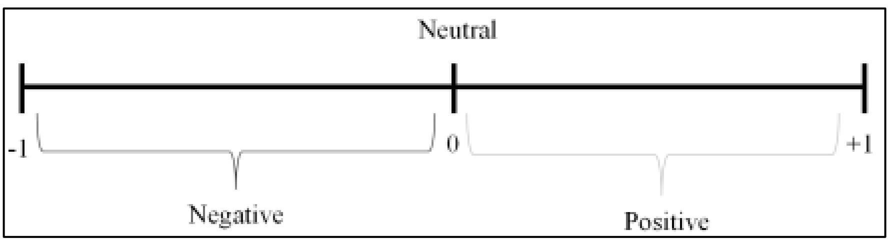

Figure 1: Sentiment compound score classification based on positive, neutral, and negative

3.4 Data Scaling and Transformation

Due to the different structure of the historical price data and sentiment data, the value to indicate the variable can be different. For example, the price would be in value RM3.5, while volume will be in quantity of stock like 150,000 shares, and sentiment score will be in +- value like 0.25. This causes an inconsistency of data across the model building and correlation determination. To solve this problem, feature scaling between both datasets is needed. A MinMaxscaler has been used to scale each feature to a range of [0,1] by using the formula below:

$$x = \frac{x - x\,min}{x\,max - x\,min} \tag{1}$$

To ensure each feature contributes equally, MinMaxScaler was applied to scale values within [0,1], preventing larger values from dominating smaller ones in the model. Additionally, sentiment data transformation was needed due to imbalance, as its availability depends on user comments, whereas historical price data follows trading days. This mismatch could cause missing values (NaN) in regression analysis. To address this, zero-value replacement was applied separately, ensuring the original sentiment scores remained unchanged, while zero indicated absent sentiment data. This preprocessing step prevents errors in model building, ensuring data consistency and improving prediction accuracy.

3.5 Correlation Analysis

Correlation analysis was used to measure the strength relationship between the historical price data and sentiment data. The sample correlation coefficient estimates the population correlation as r. The r-value represents the correlation coefficient for a random sample. The correlation coefficient (r) range is from negative 1 to positive 1, which is $-1 \leq r \leq +1$. The closer the value is to 1 or -1, the stronger the linear correlation. To interpret the strength of the relationship between sentiment score and stock price, this study adopts the correlation scale shown in Table 2, which categories the relationship from “very weak” to “very strong” based on the r value.

Table 2: Correlation Coefficient interpretation

Strength of the relationship	r
Very Weak	0.00-0.19
Weak	0.20-0.39
Moderate	0.40-0.59
Strong	0.60-0.79
Very Strong	0.80-1.00

3.6 Experiment Setting for Machine Learning Model

The focus of the stock prediction is on the closing price of the stock, and this was defined as the y target. Since this research is aimed at predicting the stock price using historical price data and sentiment data that consists of different attributes, the features selected are Open price, Highest Price, Lowest Price, Volume, and Sentiment Scores. A total of 5 features were selected as the independent variable as x.

3.6.1 Multiple Linear Regression (LR)

Linear regression is a supervised machine learning method used to model the relationship between dependent (y) and independent (x) variables by fitting a linear line to input data. Bansal *et al.* (2022) utilized linear regression for daily stock value prediction, demonstrating its effectiveness. Figure 2 shows the model setting where using the LinearRegression() function from the scikit-learn library with default parameters. The entire data was used for testing and evaluation, and predictions were generated using predict().

```
# Initialize the model
mlr = LinearRegression()
mlr.fit(X,y)

# 2. Predict on test data
MLR_pred = mlr.predict(X)
```

Figure 2: Multiple linear regression model setting

3.6.2 Support Vector Regression (SVR)

Support Vector Machine (SVM) is a supervised machine learning algorithm used for classification and regression. Support Vector Regression (SVR) finds the optimal hyperplane that maximizes the margin, ensuring future data points are classified with confidence (Sakhare and Imambi, 2019). This study applies Support Vector Regression using the SVR() function from the scikit-learn library with default settings with using the default Radial Basis Function (RBF) kernel shown in Figure 3, which is suitable for capturing nonlinear relationships in the data.

```
# Initialize and fit the model
svm = SVR()
svm.fit(X, y)

# Predict on test data
SVM_pred = svm.predict(X)
```

Figure 3: Support vector regression model setting

3.6.3 Random Forest (RF)

Random Forest is an ensemble learning method that constructs multiple decision trees during training and outputs the mean prediction of the individual trees for regression tasks to solve a complex problem (Sharma *et al.*, 2024). Figure 4 shows the model setting for Random Forest, which RandomForestRegressor() function from scikit-learn. The model was configured with random_state=42 to ensure reproducibility, while other parameters such as the number of trees (n_estimators) and maximum depth were left at default values. Model testing and prediction were conducted using. .fit(X, y) and.predict(X), respectively.

```
# Initialize and fit the model
rf = RandomForestRegressor(random_state=42)
rf.fit(X, y)

# Predict on test data
RF_pred = rf.predict(X)
```

Figure 4: Random forest model setting

3.6.4 Extreme Gradient Boost (XGBoost)

Extreme Gradient Boost, known as XGBoost, is an algorithm that further enhances or optimizes the gradient boosting technique. This technique was proposed by Chen and Guestrin (2016) with the main improvement of Gradient Boost Decision Trees (GBDT) by normalization of the loss function to mitigate model variance. From Figure 5, XGBoost was implemented using the XGBRegressor() function from the xgboost library. The model was initialized with random_state=42, while other hyperparameters were left at their default values, as n_estimators=100, learning_rate=0.3, and max_depth=6. The XGBoost can effectively handle the missing value issue and incomplete eigenvalue, which automatically figures out the splitting direction with learning and avoids over-fitting. Prediction was generated using .predict(X).

```
# Initialize and fit the model
xgb = XGBRegressor(random_state=42)
xgb.fit(X, y)

# Predict on test data
XGB_pred = xgb.predict(X)
```

Figure 5: Extreme gradient boost model setting

3.6.5 Multilayer Perceptron (MLP) Neural Network

The Multilayer Perceptron MLP is a type of neural network that consists of multiple layers of neurons. A multilayer perceptron network has three layers, which are the input layer, hidden layer, and output layer (Sun, 2025). The first layer consists of the input layer, while the last layer is the output layer, which is the target variable, and all layers between the input and output layer are called hidden layers. The model organization for MLP is shown in Figure 6, where the model was implemented using the MLPRegressor() function from the scikit-learn library. The model was initialized with random_state=42 to ensure reproducibility and max_iter=1000 to allow sufficient convergence during testing. Other parameters, such as the number of hidden neurons (default: 100), activation function (ReLU), and solver (Adam), were retained at their default values, and predictions were generated using .predict().

```
# Initialize and fit the model
nn = MLPRegressor(random_state=42, max_iter=1000)
nn.fit(X, y)

# Predict on test data
NN_pred = nn.predict(X)
```

Figure 6: Multilayer perceptron neural network model setting

3.6.6 Long Short-Term Memory (LSTM)

Long Short-Term Memory (LSTM) is a recurrent neural network (RNN) designed to learn order dependence in sequence prediction problems, making it effective for stock price prediction. It retains past information, crucial for forecasting future prices. The LSTM structure includes a memory cell, input gate, forget gate, and output gate (Shao, 2023). From Figure 7, a Long Short-Term Memory (LSTM) model was implemented using the Sequential API from the Keras library to predict stock closing prices. The input data was reshaped into a 3D structure using X.values.reshape((X.shape[0], X.shape[1], 1)) to meet the input requirements of LSTM layers. The model setting consisted of two stacked LSTM layers, each with 50 units. The first layer was configured with return_sequences=True to pass sequential output to the next LSTM layer. A Dense layer with a single neuron was added to generate the final prediction. The model was compiled using the Adam optimizer and mean squared error (MSE) as the loss function. It was trained for 50 epochs with a batch size of 32 using the entire dataset, and predictions were generated using the .predict() method.

```
# Reshaping the data for LSTM (required)
X_lstm = X.values.reshape((X.shape[0], X.shape[1], 1))

# Initialize the LSTM model
lstm_model = Sequential()
lstm_model.add(LSTM(units=50, return_sequences=True, input_shape=(X_lstm.shape[1], 1)))
lstm_model.add(LSTM(units=50))
lstm_model.add(Dense(1))

lstm_model.compile(optimizer='adam', loss='mse')

# Train the model (use the entire data for training as per your requirement)
lstm_model.fit(X_lstm, y, epochs=50, batch_size=32, verbose=0)

# Predict using the model
LSTM_pred = lstm_model.predict(X_lstm).flatten()
```

Figure 7: Long short-term memory model setting

3.7 Model Evaluation

Table 3 indicates that all of the machine model builds were evaluated using 4 metrics, which are Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Coefficient of determination R-square (R^2). For the performance metric MSE, RMSE, and MAE mean that the smaller the value, the closer the predicted value to the true value, and the lower the error. While the coefficient of determination $R^{2,}$ the closer the value to 1, the better the model fit.

Table 3: Evaluation metric formula

Evaluation Metric	Formula
Mean Squared Error	$MSE = \frac{1}{n}\sum_{i=1}^{n}(Oi - Pi)^2$
Root Mean Squared Error	$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(Oi - Pi)^2}$
Mean Absolute Error	$MAE = \frac{1}{n}\sum_{i=1}^{n}\lvert Ai - Pi\rvert$
Coefficient of Determination (R^2)	$\text{R2} = \frac{SSR}{SST} = \frac{Sum\ of\ Square\ Regression}{Total\ Sum\ of\ Square}$

4. Results and Discussion

4.1 Descriptive Analysis

The details of both the dataset's historical price data and sentiment data are shown below in Table 4. The total number of historical price data collected is 24,483. The historical price is from 28/10/2014 to 28/10/2024, for 10 years. Due to the different trading days, the amount of some companies' stock data obtained was different. This was due to the holiday, suspension of trading at some stocks, and some common acquisition requests from the company to stop trading. For sentiment data, the total sentiment data collected is 7686 data which includes the comments and reviews from the investors who commented on the application of the KLSE screener. The comment was gathered from the past year, 28/10/2023 to 28/10/2024, and was not available daily, as comments are posted at irregular intervals depending on the user activity. For each company's stock, the sentiment data depends on the user comment.

Table 4: Total dataset obtained for historical price data and sentiment data

No	Company Stock	Number of Historical Price Dataset	Number of Sentiment dataset
1	Celcom Digi Berhad	2448	173
2	CIMB Group Holding Berhad	2452	370
3	Genting Berhad	2452	1846
4	MayBank Berhad	2452	1184
5	Nestle Berhad	2435	255
6	Petronas Chemical Group Berhad	2452	372
7	Press Metal Aluminium Holding Berhad	2438	114
8	Public Bank Berhad	2450	1232
9	Tenaga Nasional Berhad	2452	482
10	YTL Corporation Berhad	2452	1658
	Total	24483	7686

4.2 Sentiment Analysis

In this research, the sentiment identification of the review was done by using the VADER approach. There were three classifications of the review were identified, which are positive, negative, and neutral sentiment. Table 5 shows an example of the review with its compound value by the VADER. The compound score was computed from the polarity score of each word, and then compounded it became the sentiment to show the sensitivity of the review.

Table 5: Sentiment analysis result

No	Review after cleaning	Sentiment Score	Sentiment Classification
1	buy great glorious your set today give greater profit dare go	Compound: 0.9477	Positive
2	delay qr announc bad share price sentiment mean damn bad bad news	Compound: -0.9231	Negative
3	resort world la vega money launder case solv yet	Compound: 0.0	Neutral

There are a total of 10 companies whose stocks have been selected to conduct the sentiment analysis on the investor review at the KLSE Screener application. Based on Table 6, a total of 7686 valid reviews were analysed. The result in Table 6 shows that the majority of the reviews are positive, with 3835 reviews having positive sentiment. This means that most investors anticipate the stock value will increase in the future. For negative sentiment, there are 1470 negative sentiments, which represent the perception of the investors seen the stock value is likely to decrease in the future. Among the sentiment analysis, there is a neutral sentiment that consists of 2381 reviews. For neutral sentiment results in this research did not provide any significant contribution to the prediction of stock price during the machine learning model building. This is because the neutral sentiment results only play a role in classifying the review as having positive or negative polarity, or even do not have any sentiment indicated, which will affect the accuracy of the sentiment analysis. To avoid uncertainty of text sentiment classification, neutral is used for classifying the neutral opinion or text that does not have a key role in the positive or negative category. In this study, the main contribution finding is based on the positive and negative sentiment.

Table 6: Overview of sentiment analysis results

No	Stock	Total review	Positive	Neutral	Negative
1	Celcom Digi Berhad	173	75	53	45
2	CIMB Group Holding Berhad	370	220	110	40
3	Genting Berhad	1846	914	529	403
4	MayBank Berhad	1184	643	375	166
5	Nestle Berhad	255	124	67	64
6	Petronas Chemical Group Berhad	372	189	102	81
7	Press Metal Aluminium Holding Berhad	114	56	37	21
8	Public Bank	1232	591	377	264
9	Tenaga Nasional Berhad	482	239	165	78
10	YTL Corporation Berhad	1658	784	566	308
	Total	7686	3835	2381	1470

4.3 Correlation Analysis

The correlation value is shown in Table 7 for each sentiment score with the historical price data, such as close price, open price, highest price, lowest price, and volume. In this study, the correlation between sentiment data and historical price data can be explained. The positive relationship shows that when the price increases, the emotion of investor will be optimistic (positive). In contrast, when the price decreases, the investor will react with pessimistic emotions. For negative relationship between sentiment data and historical price data shows that when a price tends to increase, the emotions of investors will show pessimistic, while when a price tends to decrease, the investor looks optimistic.

Based on the strength of the correlation relationship, for companies, CIMB Group Holding Berhad and Press Metal Aluminium Berhad show a weak positive correlation relationship between the sentiment data and historical price data. While for others, company stocks show a weak negative relationship between the sentiment data and historical price data. Although many of the correlations appear weak in isolation, they may still provide predictive value when combined with other features. Such subtle linkages and interactions can be exploited by machine learning models, particularly non-linear and ensemble techniques, to enhance classification performance.

Among the results, the correlation between the sentiment score and volume showed a different correlation when compared with the price data. For stock Genting Berhad and Nestle Berhad, the correlation between the sentiment score and price was a negative relationship, but for volume, it shows a weak positive relationship with the sentiment score. Overall, for the correlation result, most of the company stock shows a negative relationship with the sentiment data and price data, while only two company stocks show a positive relationship with the sentiment data and price data. Table 7 shows the result of the correlation between the variables for both datasets.

Table 7: Correlation between sentiment score and historical price data

	Correlation (r value)	Sentiment Score				
No	Company Stock	Close price	Open Price	Highest Price	Lowest Price	Volume
1	Celcom Digi Berhad	-0.0475	-0.0475	-0.0471	-0.0466	-0.0158
2	CIMB Group Holding Berhad	0.2148	0.2152	0.2157	0.2162	0.0823
3	Genting Berhad	-0.0692	-0.0689	-0.0692	-0.0690	0.023
4	MayBank Berhad	-0.0608	-0.0601	-0.0626	-0.0589	-0.04
5	Nestle Berhad	-0.0085	-0.0086	-0.0080	-0.0093	0.0124
6	Petronas Chemical Group Berhad	-0.0746	-0.0713	-0.0729	-0.0741	-0.0492
7	Press Metal Aluminium Holding Berhad	0.0795	0.0806	0.0802	0.0809	0.0278
8	Public Bank Berhad	-0.1989	-0.1989	-0.1988	-0.1989	0.1391
9	Tenaga Nasional Berhad	-0.1012	-0.1026	-0.1020	-0.1021	-0.0022
10	YTL Corporation Berhad	-0.0934	-0.0957	-0.0932	-0.0953	0.003

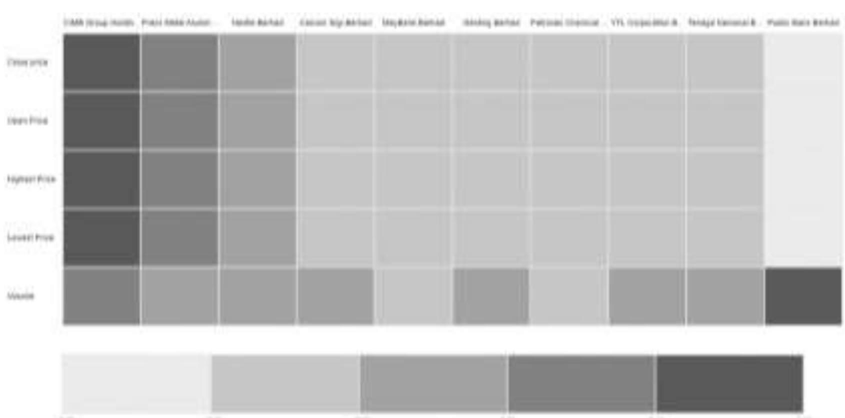

Figure 8: Heatmap for correlation analysis between sentiment score and historical price data

4.4 Model Evaluation Result

4.4.1 Mean Squared Error (MSE)

Table 8 shows the summarized results of the 6 machine learning models for 10 company stocks based on the evaluation metric MSE. To show which machine learning model performs with the least error among the models, a ranking of 1- 6 has been used to indicate which machine learning model has the best performance metric, where a rank of 6 shows the models that have the smallest error within all the models, and company stock which rank of 1 shows the models that have the highest error.

Table 8: Summarize Machine Learning Performance based on MSE

		Mean Squared Error (MSE)					
No	Company Stock	LR	RF	SVM	XGBoost	MLP	LSTM
1	Celcom Digi Berhad	0.00007	0.00001	0.00124	0.00002	0.00020	0.00017
2	CIMB Group Holding Berhad	0.00003	0.00001	0.00097	0.00001	0.00019	0.00017
3	Genting Berhad	0.00003	0.00001	0.00092	0.00001	0.00025	0.00012
4	MayBank Berhad	0.00007	0.00002	0.00040	0.00001	0.00026	0.00024
5	Nestle Berhad	0.00004	0.00001	0.00135	0.00001	0.00019	0.00009
6	Petronas Chemical Group Berhad	0.00007	0.00001	0.00139	0.00001	0.00022	0.00023
7	Press Metal Aluminium Holding Berhad	0.00002	0.00004	0.00071	0.00001	0.00018	0.00011
8	Public Bank Berhad	0.00001	0.00001	0.00215	0.00001	0.00013	0.00004
9	Tenaga Nasional Berhad	0.00007	0.00001	0.00118	0.00001	0.00025	0.00019
10	YTL Corporation Berhad	0.00002	0.00005	0.00221	0.00001	0.00010	0.00005
	Rank	4	5	1	6	2	3

From Table 8, the MSE value obtained for XGBoost was ranked 6 among the models, which means that among the 6 models built XGBoost model has the smallest error rate. This indicates the prediction of the stock price is more accurate compared with other models. This shows that XGBoost effectively captures a complex pattern between the historical price data and sentiment. In contrast, the SVR has the worst MSE value, which means that the error is higher than other models, and this may be due to the SVR's sensitivity to noisy or sparse sentiment data and limiting its ability to predict stock price using sentiment data.

4.4.2 Root Mean Squared Error

For RMSE, the square root of MSE, the lower the RMSE, the better the model performs (Table 9). This metric shows a similar result compared with the MSE metric. The model XGBoost shows the lowest error among the models, which ranks number 6 among the models that performed well, with the low value of RMSE corresponding to the higher model performance. A similar result from MSE and RMSE demonstrates its effectiveness in handling the complex relationship between the different pattern datasets. However, SVR and MLR show comparatively higher RMSE values, showing their limitations in predicting non-linear patterns and irregular sentiment data.

Table 9: Summarize Machine Learning Performance based on RMSE

		Root Mean Squared Error (RMSE)					
	Company Stock	LR	RF	SVM	XGBoost	MLP	LSTM
1	Celcom Digi Berhad	0.00863	0.00368	0.03527	0.00407	0.01431	0.01298
2	CIMB Group Holding Berhad	0.00541	0.00267	0.03109	0.00257	0.01375	0.01320
3	Genting Berhad	0.00565	0.00270	0.03037	0.00268	0.01592	0.01106
4	MayBank Berhad	0.00826	0.00389	0.02002	0.00347	0.01622	0.01558
5	Nestle Berhad	0.00633	0.00279	0.03680	0.00254	0.01389	0.00954
6	Petronas Chemical Group Berhad	0.00812	0.00353	0.03724	0.00348	0.01475	0.01507
7	Press Metal Aluminium Holding Berhad	0.00467	0.00211	0.02662	0.00245	0.01337	0.01063
8	Public Bank Berhad	0.00226	0.00105	0.04640	0.00150	0.01158	0.00668
9	Tenaga Nasional Berhad	0.00841	0.00334	0.03433	0.00309	0.01566	0.01379
10	YTL Corporation Berhad	0.00388	0.00214	0.04702	0.00204	0.00998	0.00740
	Rank	4	5	1	6	2	3

4.4.3 Mean Absolute Error (MAE)

For the MAE metric (Table 10), there is some difference in the performance result when compared with the previous two metrics, MSE and RMSE. The best model predicted the stock price is the RF model, which ranks 6, followed by XGBoost, ranking 5th for the performance evaluation. This shows that XGBoost may be more sensitive to minimizing larger errors as captured by MSE and RMSE, while the Random Forest model provides more consistent predictions across all data and resulting in smaller average deviation. In contrast, the worst performance model is also ranked by SVR compared to the other models. This shows a low capability to deal with non-linear data such as investor sentiment.

Table 10: Summarize Machine Learning Performance based on MAE

		Mean Absolute Error (MAE)					
	Company Stock	LR	RF	SVM	XGBoost	MLP	LSTM
1	Celcom Digi Berhad	0.00649	0.00264	0.02892	0.00305	0.01071	0.00930
2	CIMB Group Holding Berhad	0.00397	0.00178	0.02652	0.00197	0.00951	0.01009
3	Genting Berhad	0.00413	0.00187	0.02488	0.00195	0.01147	0.00819
4	MayBank Berhad	0.00606	0.00259	0.01278	0.00264	0.01134	0.01159
5	Nestle Berhad	0.00396	0.00163	0.03102	0.00185	0.01050	0.00564
6	Petronas Chemical Group Berhad	0.00588	0.00242	0.03243	0.00263	0.01109	0.01065
7	Press Metal Aluminium Holding Berhad	0.00325	0.00142	0.01862	0.00182	0.00947	0.00799
8	Public Bank Berhad	0.00133	0.00057	0.03748	0.00102	0.00830	0.00479
9	Tenaga Nasional Berhad	0.00542	0.00221	0.02940	0.00229	0.01162	0.01033
10	YTL Corporation Berhad	0.00258	0.00120	0.04042	0.00153	0.00670	0.00430
	Rank	4	6	1	5	2	3

4.4.4 R-square (R^2)

R-square has been ranked based on the model fit, which means the higher the value closer to 1, the higher the rank will give to the model. From Table 11, the overall R-square among the models shows more than 0.90 or 90% which indicates that 90% of the predictor or the independent variable can explain the variance in the predicted stock price. This indicates the model provides a good fit to the data. XGBoost shows the highest R-square, which ranked 6th. For RF, the performance metric for R-squared also ranked number 5 among the model builds. These show that the strength of the ensemble learning techniques like XGBoost and Random Forest in capturing the complex and non-linear relationship in stock market prediction, during combining historical price with sentiment feature. The worst performance model was also determined by SVR, which ranked 1 in the least R-square which consistent with weaker performance across other metrics.

Table 11: Summarize Machine Learning Performance based on R-square (R^2)

		R-square (R^2)					
	Company Stock	LR	RF	SVM	XGBoost	MLP	LSTM
1	Celcom Digi Berhad	0.99790	0.99962	0.96489	0.99953	0.99422	0.99524
2	CIMB Group Holding Berhad	0.99907	0.99977	0.96Ea927	0.99979	0.99399	0.99446
3	Genting Berhad	0.99959	0.99991	0.98829	0.99991	0.99678	0.99845
4	MayBank Berhad	0.99814	0.99959	0.98910	0.99967	0.99285	0.99340
5	Nestle Berhad	0.99960	0.99993	0.98665	0.99994	0.99810	0.99910
6	Petronas Chemical Group Berhad	0.99817	0.99965	0.96141	0.99966	0.99395	0.99368
7	Press Metal Aluminium Holding Berhad	0.99967	0.99993	0.98935	0.99991	0.99732	0.99830
8	Public Bank Berhad	0.99996	0.99999	0.98400	0.99998	0.99900	0.99967
9	Tenaga Nasional Berhad	0.99901	0.99985	0.98353	0.99987	0.99657	0.99734
10	YTL Corporation Berhad	0.99959	0.99987	0.93910	0.99988	0.99726	0.99849
	Rank	4	5	1	6	2	3

4.5 Impact of the Correlation Relationship Between Historical Price and Sentiment Data

The study findings reveal varying correlation relationships between historical stock prices and sentiment data, emphasizing that investor emotions and expressions differ across stocks. Notably, CIMB Group Holding Berhad and Press Metal Aluminium Berhad exhibited a weak positive correlation (r=0.21), indicating that investors tend to be optimistic and buy stocks during price increases, while pessimism leads to selling pressure during declines. This aligns with Low *et al.* (2017), who found a moderate positive correlation (0.5841) between financial news sentiment and stock prices, highlighting the influence of media tone on market movements. Similarly, Ahmed *et al.* (2024) identified a moderate correlation between historical prices and sentiment scores.

However, for several stocks, the correlation has resulted in negative sentiment, which is either weakly negative or a near-zero relation, and this indicates that positive sentiment did not correspond with the price increase. One explanation is that the investor sentiment results in this study tend to reflect post-event reaction, which the investor sentiment expresses through the review and comment is not forward-looking but instead of reflecting to the current event after a significant price movement has already occurred. Example, when the company announce of strong profit earning the price have already rise and lead to overvaluation which the price has risen beyond the value that investor is expected and continue buying the stock may lead to loss because the investor who bought earlier may sell to get profit and demand reduce will cause price movement where the new investor that hesitate to enter the market at high prices will fearing. This situation reflects contrarian investor behaviour, which expects future market reversals was present as some investors buy during pessimism and sell during optimism that showing an inverse of expected correlation. The result underscores that sentiment is not always a predictive signal, but rather one of the many behavioural and market-driven factors influencing stock prices.

An important consideration is whether sentiment precedes or follows price movements. The results of this study suggest that sentiment often lags behind price changes, as many investor comments appear to be reactions to events that have already affected stock prices, such as earnings announcements or sudden price swings. This reactive pattern indicates that sentiment reflects prevailing market conditions rather than consistently providing a forward-looking signal. However, in certain cases, contrarian behavior emerged, where negative sentiment coincided with rising prices or optimism surfaced during price declines, suggesting that sentiment can occasionally act as an anticipatory signal. These mixed findings highlight that the role of sentiment is context-dependent, more frequently reflective but at times predictive of subsequent market movements.

4.6 Comparison of Machine Learning Model to Predict the Stock Price

This study evaluated six machine learning models for stock price prediction using historical and sentiment data, with performance measured using MSE, RMSE, MAE, and R^2. The ranking method determined that Extreme Gradient Boost (XGBoost) was the best model, achieving 23 total ranking points, followed by Random Forest (21), Multiple Linear Regression (16), Long Short-Term Memory (LSTM) (12), Multilayer Perceptron Neural Network (8), and Support Vector Regression (SVR) (4) as the least effective model.

Table 12: Summarize Machine Learning Performance based on R-square (R2)

		Machine Learning Model					
	Performance Metric	LR	RF	SVM	XGBoost	MLP	LSTM
1	MSE	4	5	1	6	2	3
2	RMSE	4	5	1	6	2	3
3	MAE	4	6	1	5	2	3
4	R^2	4	5	1	6	2	3
	Total Number of Rank	16	21	4	23	8	12

XGBoost demonstrated superior predictive accuracy due to its ability to handle complex relationships between historical price and sentiment data, aligning with Sonkavde *et al.* (2023), who found XGBoost outperformed other models. Additionally, ensemble models combining Random Forest, XGBoost, and LSTM further improved prediction performance. While prior research, such as De Lange (2023), identified LSTM and SVM as highly accurate for stock price prediction, sentiment data consistently enhanced prediction accuracy. Al Ridhawi (2021) also reported a 17% accuracy improvement when incorporating sentiment analysis. This study highlights that different stocks exhibit varying error metrics, emphasizing that price and sentiment fluctuations impact model performance. These findings provide valuable insights for investors, researchers, and financial analysts navigating stock market complexities, volatility, and dynamic trends.

5. Conclusion and Future Research

5.1 Summary

The usage of the KLSE Screener comments as a localised sentiment source, in contrast to previous Malaysian market prediction studies that mostly used financial ratios, news stories, or worldwide sentiment databases. This is a unique addition since the remarks provide context-specific insights into market behaviour and represent the real-time viewpoints of Malaysian retail investors. This research focuses on predicting the Bursa Malaysia KLCI index stock market by integrating historical price data and sentiment data for 10 selected companies. Among the companies, English-based content still predominates, despite the fact that they offer comments in a variety of languages and slang. Because VADER is robust to capitalisation, punctuation, and colloquial idioms, it was selected for brief, informal texts like social media postings and online reviews. Moreover, six machine learning models, Multiple Linear Regression, Random Forest, Support Vector Regression, Extreme Gradient

Boost (XGBoost), Multilayer Perceptron Neural Network, and Long Short-Term Memory (LSTM) were implemented to determine the best-performing model. The models were evaluated using Mean Squared Error (MSE), Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and R-squared (R^2).

Additionally, this research examined correlation relationships between historical price and sentiment data using the Pearson Correlation Coefficient. Findings revealed both positive and negative correlations, indicating different investor behaviors. A positive correlation suggests stock prices rise with positive sentiment due to increased buying demand, whereas a negative correlation implies prices drop despite positive sentiment due to selling pressure. The study successfully identified XGBoost as the best model for stock price prediction, achieving the smallest error rate and highest ranking. These findings contribute to a better understanding of investor sentiment, stock market dynamics, and predictive modeling, aiding investors in making informed decisions.

5.2 Limitations

This study has limitations in sentiment analysis due to the use of VADER, which misclassifies some comments and reviews. While VADER is effective for English text, it struggles with Malay, Chinese, and local Malaysian languages, leading to incorrect sentiment classification. Many comments from the KLSE Screener contain mixed languages, slang, informal spelling, and domain-specific vocabulary, which VADER does not recognize, often categorizing them as neutral sentiment. This reduces the accuracy and contribution of sentiment data in the machine-learning model. The limitation highlights the need for a more localized sentiment analysis tool to improve stock market prediction in Malaysia's diverse linguistic context.

This study also faces a dataset size limitation due to the imbalance between historical price data and sentiment data from the KLSE Screener. In contrast to ten years of historical pricing data, the sentiment dataset only spans one year. The model's capacity to represent longer-term relationships between mood and price movements may have been limited by this imbalance. The differing data volumes also create an imbalance issue during machine learning testing. To address this, missing values were replaced with zero, representing neutral sentiment (no price change). However, this replacement may distort the relationship between sentiment and price. Despite efforts to manage this issue, it remains unavoidable due to the inherent differences in data characteristics. Uneven distribution of sentiment data across the 10-stock dataset also presents a limitation for stocks that have fewer reviews. This causes less representation of actual investor sentiment, leading to potential noise in the model. This limitation highlights the need for more balanced datasets to improve stock market prediction accuracy.

Another limitation arises from the short time span of sentiment data compared to the ten-year historical stock dataset. Collecting sentiment data over multiple years would help improve the temporal balance with historical price data, allowing for a more comprehensive alignment between investor behaviour and market performance.

5.3 Recommendations

Future researchers are encouraged to explore alternative sentiment classification methods such as BERT and Multilingual-BERT, which support multiple languages. However, Malaysia’s diverse languages and dialects may still lead to uninterpretable sentiment analysis. A customized VADER lexicon with Malay-specific vocabulary, idioms, and slang could improve accuracy, though expert input would be required to assign polarity scores.

To address dataset imbalance, future studies should increase dataset size, ensuring a more balanced distribution of positive and negative sentiment scores. A larger dataset allows models to learn better patterns, improving prediction accuracy. Additionally, researchers should explore advanced machine learning techniques designed to handle imbalanced and unstructured data, further enhancing stock market prediction performance. These improvements could provide more reliable insights for financial forecasting in Malaysia's unique linguistic and economic context.

Additionally, future research should consider collecting sentiment data over longer periods, ideally spanning multiple years, to match the coverage of historical stock data. This approach would provide richer insights into long-term investor sentiment trends and is expected to enhance the robustness and predictive performance of machine learning models.

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